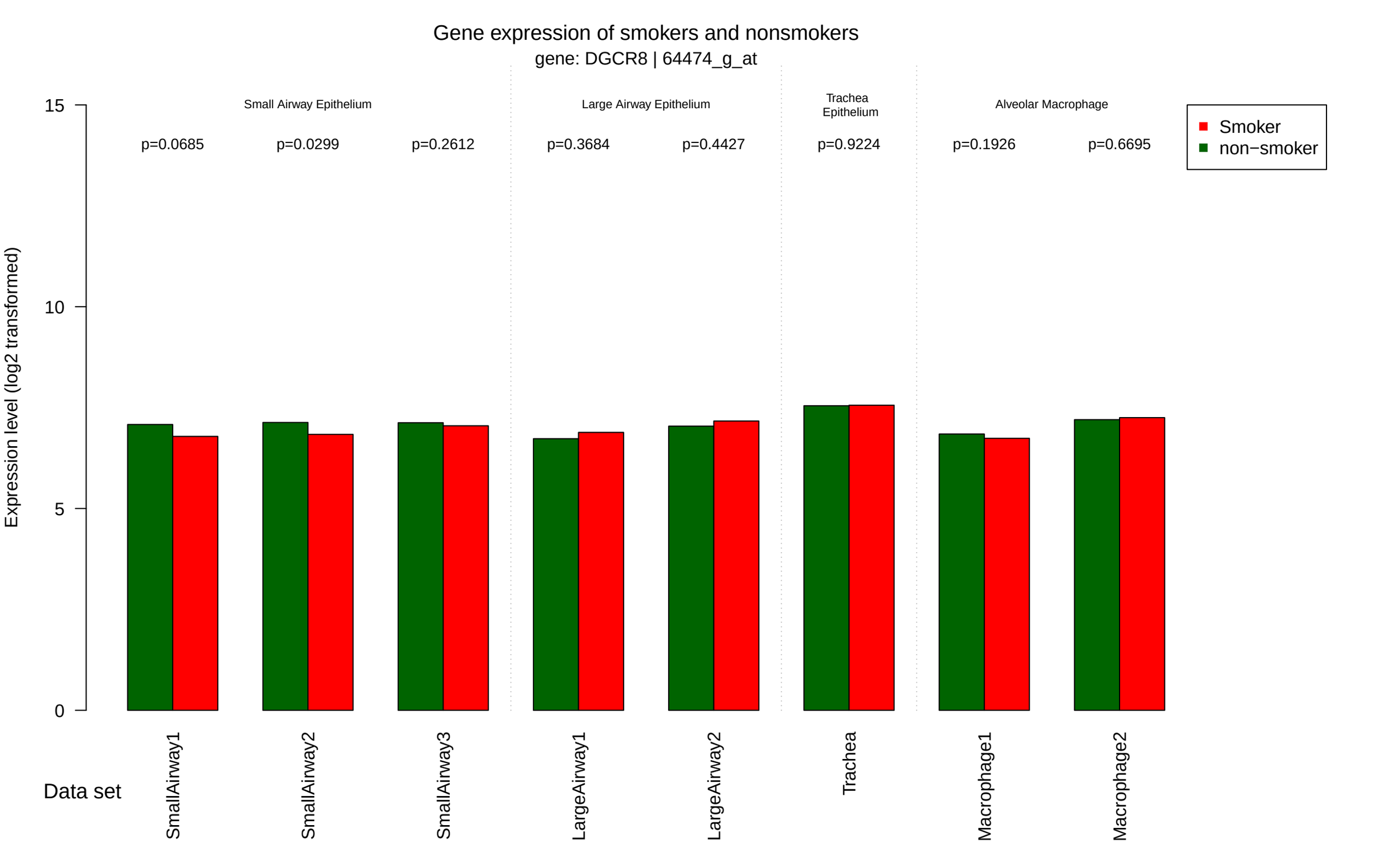
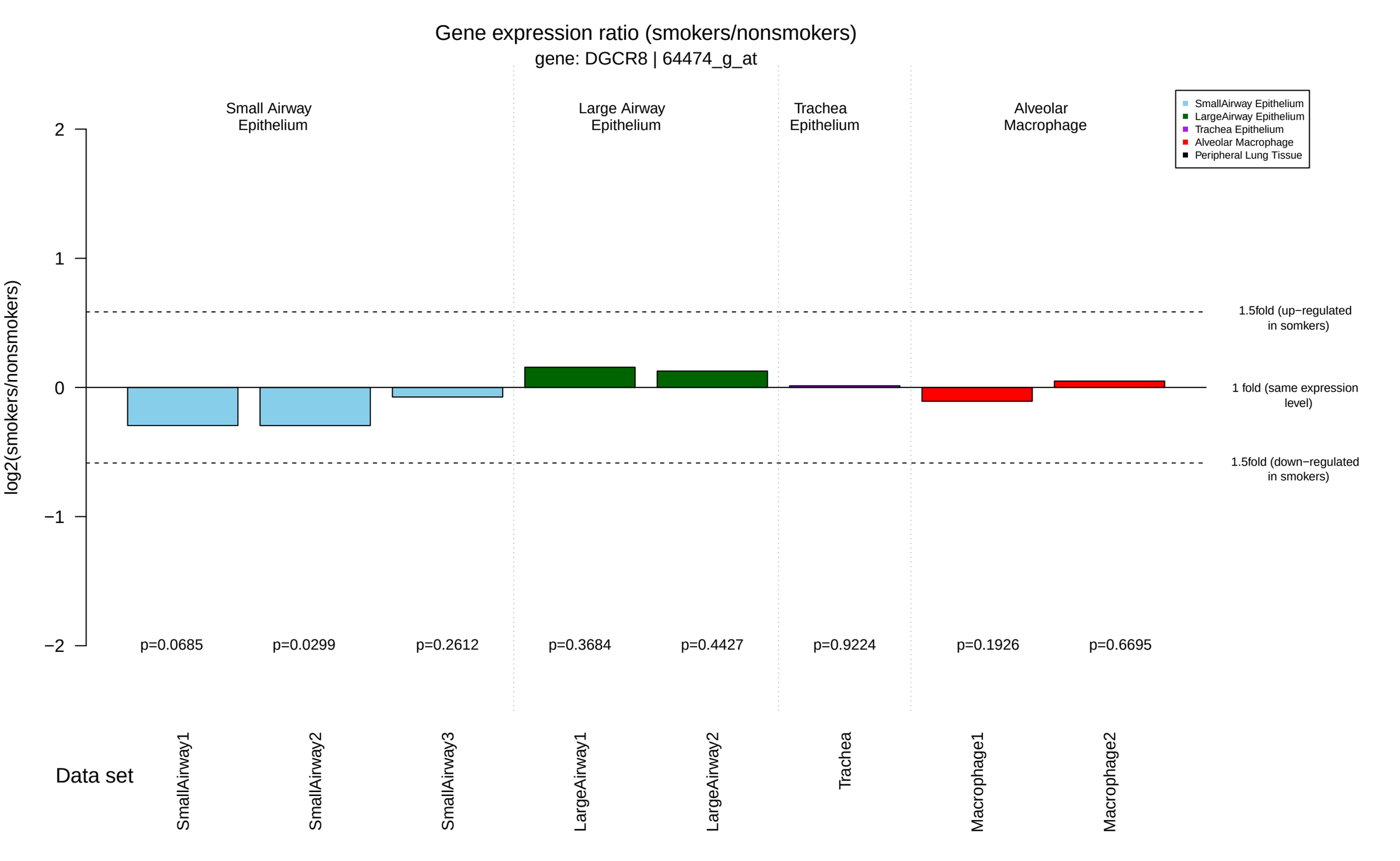
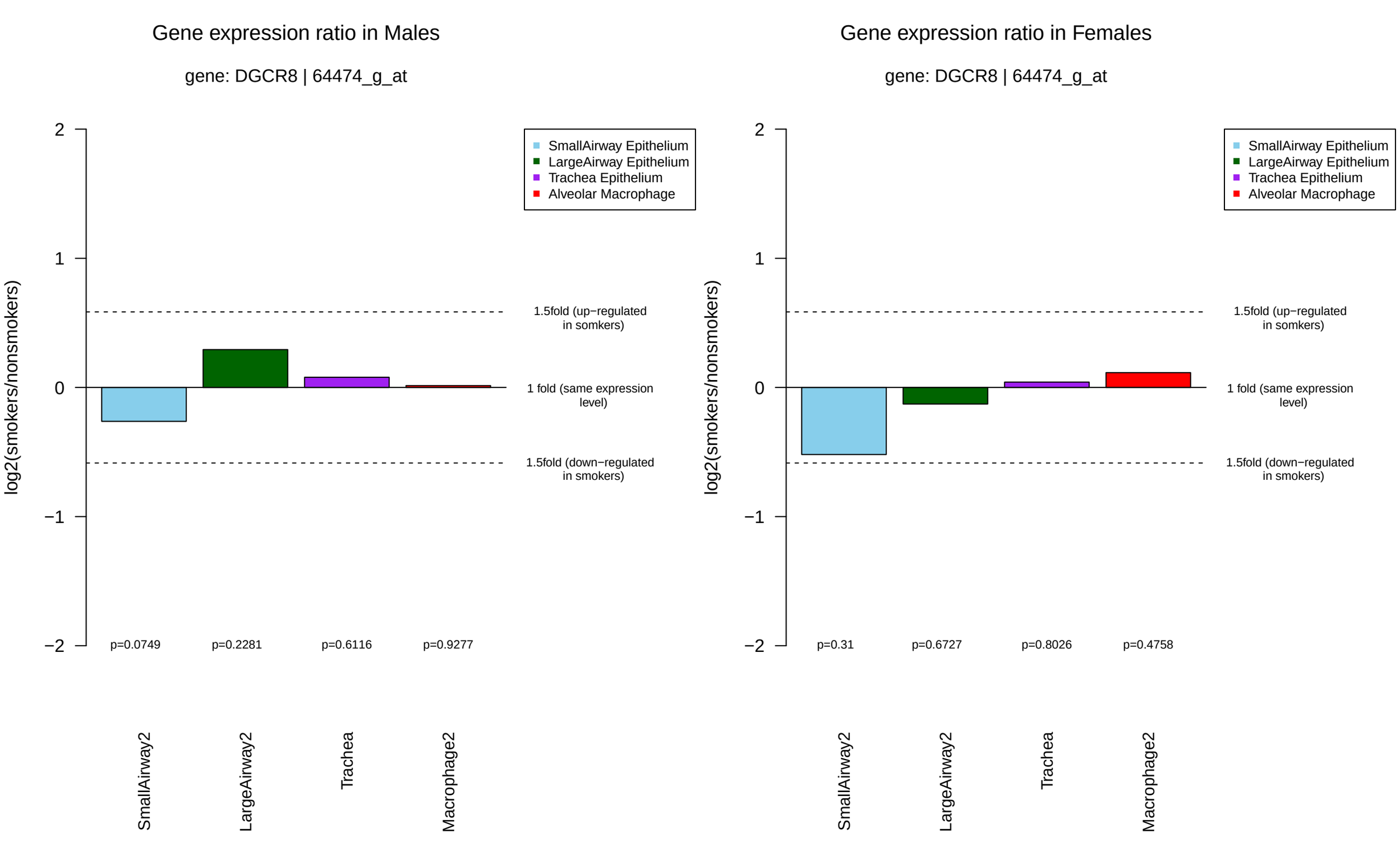
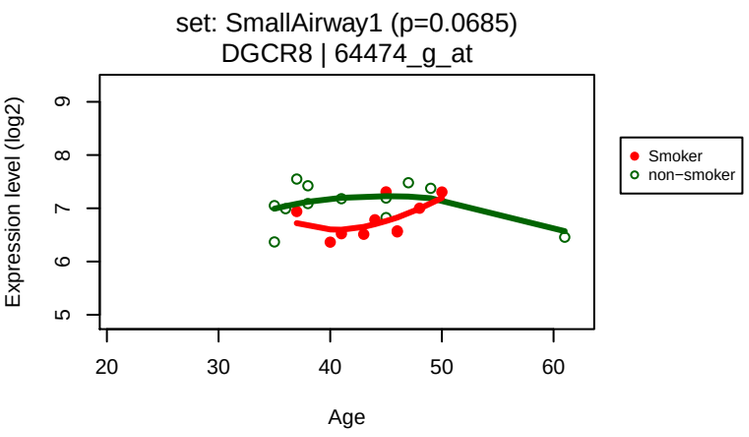
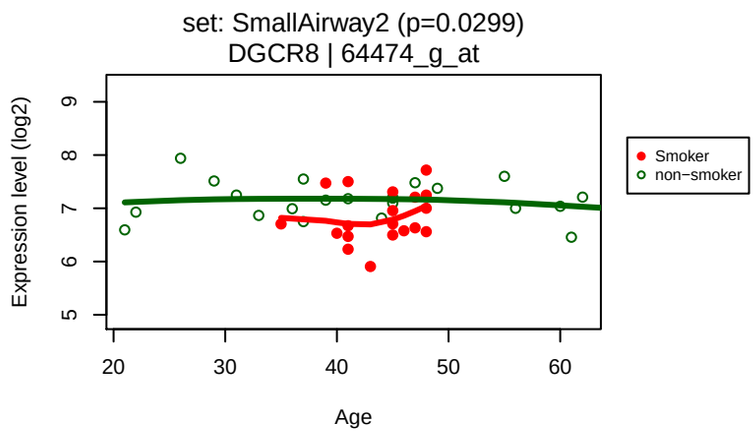
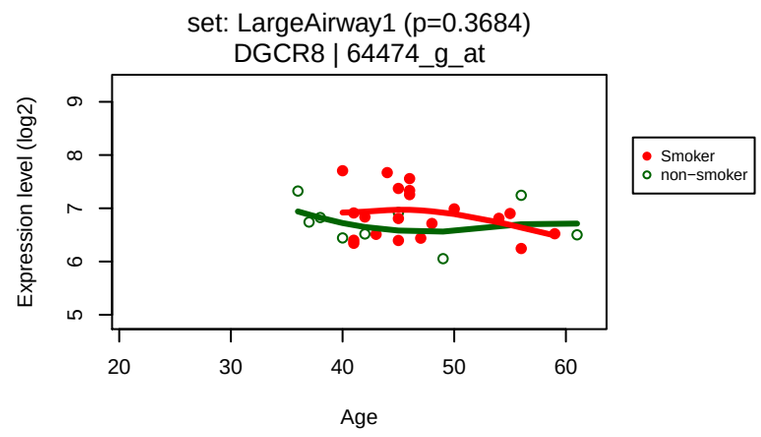

Gene expression of smokers and nonsmokers
gene: DGCR8 | 64474_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0685
p=0.0299
p=0.2612
p=0.3684
p=0.4427
p=0.9224
p=0.1926
p=0.6695
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

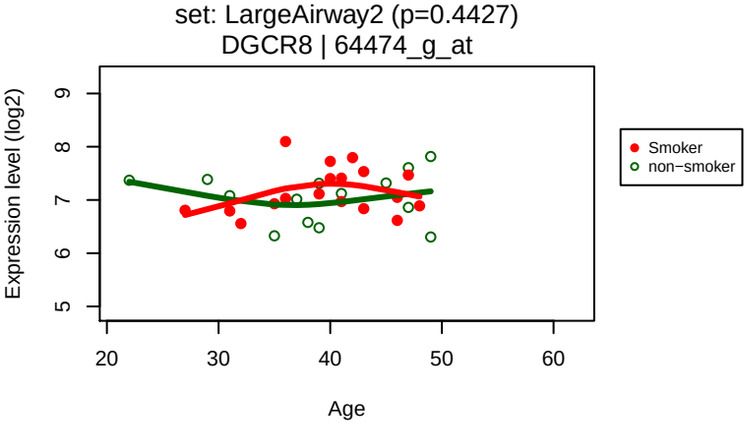

Gene expression ratio (smokers/nonsmokers)
gene: DGCR8 | 64474_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0685
p=0.0299
p=0.2612
p=0.3684
p=0.4427
p=0.9224
p=0.1926
p=0.6695
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

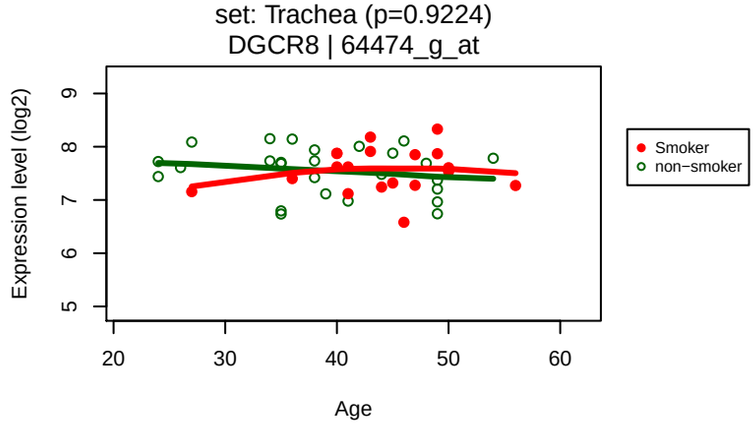

Gene expression ratio in Males
gene: DGCR8 | 64474_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0749
p=0.2281
p=0.6116
p=0.9277
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DGCR8 | 64474_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.31
p=0.6727
p=0.8026
p=0.4758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

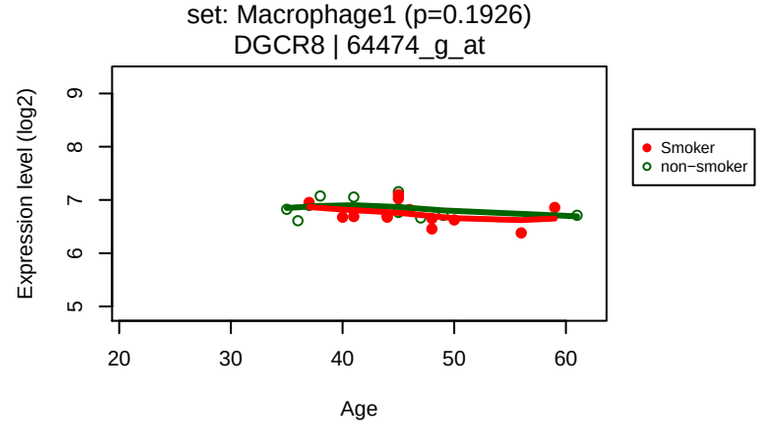

set: SmallAirway1 (p=0.0685)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker

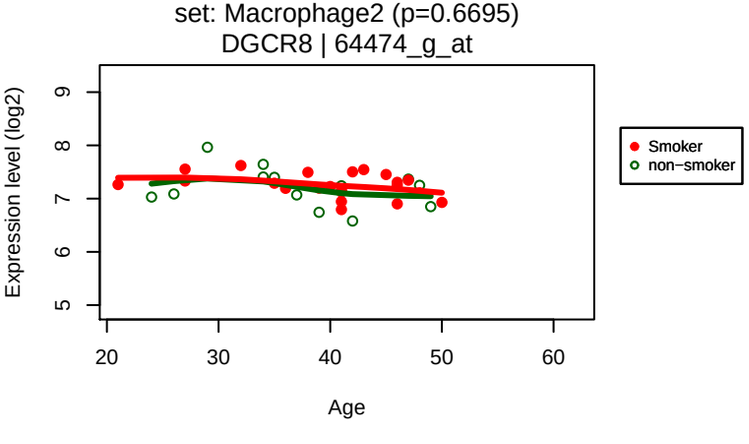

set: SmallAirway2 (p=0.0299)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker

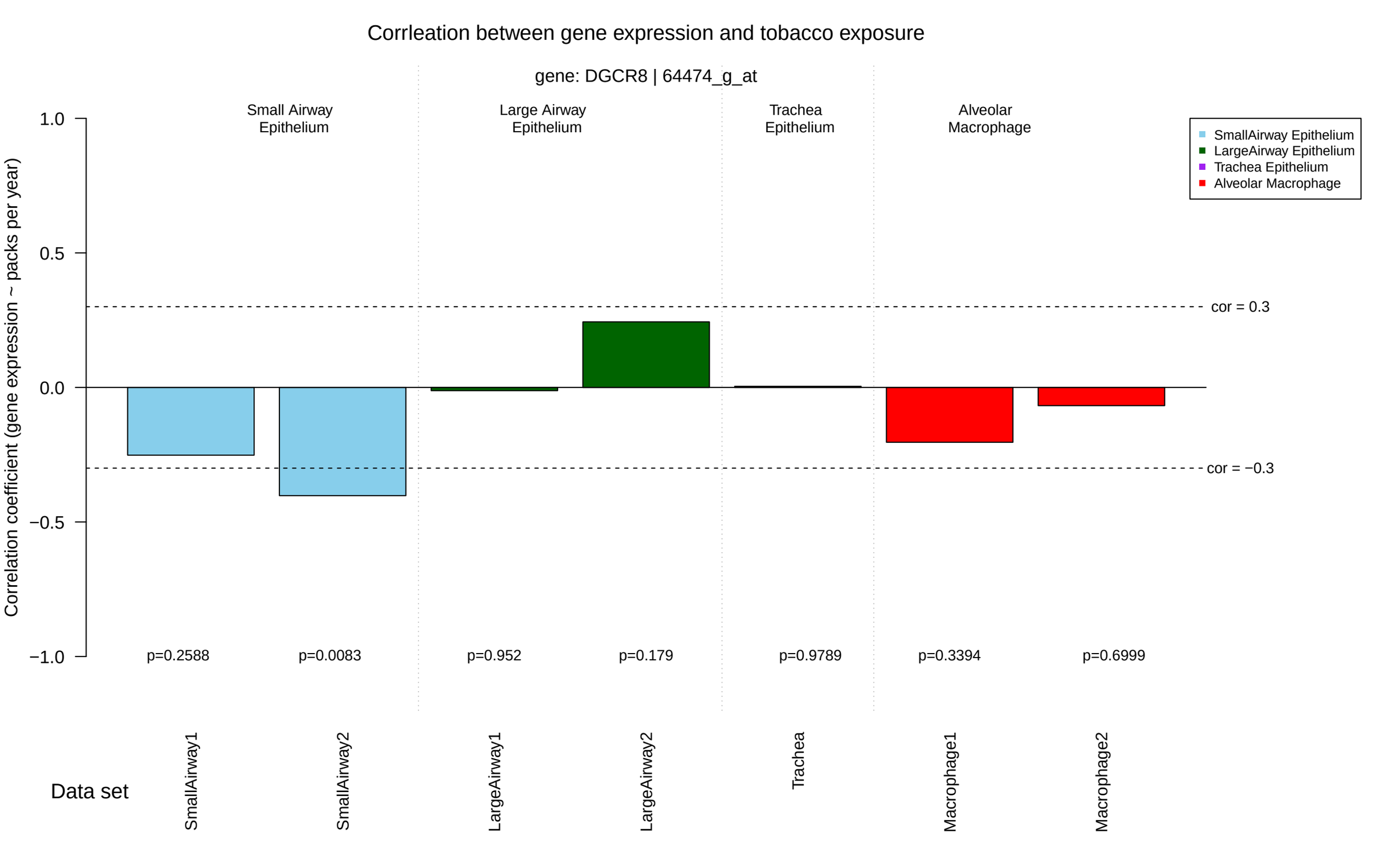

set: LargeAirway1 (p=0.3684)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4427)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9224)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1926)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6695)
DGCR8 | 64474_g_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DGCR8 | 64474_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2588
p=0.0083
p=0.952
p=0.179
p=0.9789
p=0.3394
p=0.6999
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2